

Mantavya Maan

B.S. Mathematics | Indian Institute of Technology Bombay
maanmantavya.iitb@gmail.com | +91 73074 15200 | Portfolio | LinkedIn | GitHub

Education

Indian Institute of Technology Bombay CPI: 8.86/10
B.S. in Mathematics; Minor in Game Theory and Logic, Department of Computer Science and Engineering Expected May 2027

Scholastic Achievements

- Secured top **0.49** percentile in **JEE Main 2023** examination out of **1.1 million** eligible candidates all over India.
- Among the top **0.97** percentile in **JEE Advanced 2023** examination out of **0.18 million** candidates all over India.

Professional Experience

- Palpx** | Summer Intern, Mathematical Modeling & Artificial Intelligence May 2026 – Jul 2026
- Formulated **4 predictive scoring models** via **statistical inference**, probabilistic modeling, and mathematical optimization.
 - Applied **lambda calculus**, computational logic, functional modeling, and **Retrieval-Augmented Generation** to **AI systems**.
 - Architected an **LLM orchestration platform** supporting **100+ AI models** using Bayesian inference, utility scoring, and routing.
- Gradpipe** | Machine Learning & Data Engineering Intern Nov 2025 – Dec 2025
- Architected a **high-dimensional talent discovery engine** using **vector databases** and hybrid retrieval for semantic matching.
 - Engineered an **AST-based analysis pipeline** evaluating complexity, concurrency, and memory utilization for repository scoring.
 - Designed an **LLM-as-a-Judge** evaluation framework using standardized benchmarks, reproducible scoring, and model validation.
- Fino Payments Bank** | Summer Intern, Data Analytics Jun 2025 – Jul 2025
- Analyzed **1M+ financial records** using **SQL**, **Pandas**, and **NumPy** to identify market trends and evolving customer behavior.
 - Developed interactive **Tableau dashboards** tracking **20+ KPIs** to accelerate executive decision-making and business reporting.
 - Collaborated across cross-functional teams to strengthen **data integrity**, reporting pipelines, and business analytics processes.

Key Projects

- Neural Gateway** | AI Infrastructure & Multi-LLM Orchestration May 2026 – Jul 2026
- Architected an **AI orchestration platform** routing **100+ LLMs** using Bayesian inference, Thompson sampling, and planning.
 - Designed a **KNN vector-embedding parser** using FastEmbed, ONNX, semantic retrieval, caching, and sub-**10 ms** inference.
 - Engineered a **FastAPI control plane** with OpenRouter synchronization, dynamic model registry, and adaptive policy enforcement.
 - Developed **multi-stage execution planning** supporting verifier selection, fallback routing, response streaming, and audit trails.
- Multi-Agent Orchestrator** | AI Workflow Engine Jun 2026 – Jul 2026
- Architected a coordinator-driven **multi-agent platform** with autonomous planning, shared memory, and graph-based execution.
 - Engineered an **event-driven orchestration runtime** using **LangGraph**, FastAPI, structured outputs, robust state persistence.
 - Designed a **capability-aware agent framework** enabling tool orchestration, adaptive scheduling, and reliable execution recovery.
- Prompt Optimizer** | Autonomous LLMops Platform May 2026 – Jun 2026
- Architected an **autonomous LLMops platform** for prompt optimization using genetic algorithms and deterministic evaluation.
 - Engineered a **local-first optimization engine** supporting prompt versioning, A/B deployment, SQLite persistence, and FastAPI.
 - Developed **self-improving prompt workflows** through few-shot discovery, active feedback, and continuous background evolution.
- Lead Score Model** | Mathematical Modeling and Decision Intelligence May 2026 – Jun 2026
- Built a mathematical lead qualification framework spanning over **10 scoring parameters** and **125+ CRM business variables**.
 - Applied **Bayesian inference**, Beta confidence intervals, nonlinear optimization, and probabilistic modeling for robust lead ranking.
 - Designed explainable lead prioritization using **sigmoid transforms**, exponential decay, adaptive score calibration, and optimization.

Technical Skills

Languages & Frameworks: Python, C, C++, Java, SQL, JavaScript, FastAPI, Streamlit, LangChain, Ollama, LLM Orchestration
AI Infrastructure: LLMs, RAG, Prompt Engineering, Semantic Retrieval, Vector Databases, ChromaDB, Pinecone, Vector Search
Libraries & Tools: NumPy, Pandas, SciPy, scikit-learn, Matplotlib, Git, GitHub, Jupyter Notebook, Tableau, OOP, SQLite, L^AT_EX
Mathematical Concepts: Bayesian Inference, Thompson Sampling, Statistical Inference, Nonlinear Optimization, Semantic Parsing

Leadership & Activities

- Won **silver** in **Powerlifting** at the General Sports Championship, IITB and **bronze** at the Overall Institute Powerlifting Opens.
- Coordinated **15+ T20 matches** during Aahvan and served as Hostel **Secretary**, representing over **1200+ hostel residents**.
- Completed Summer of Science in **Data Structures & Algorithms** and Season of Code in **Competitive Programming**.
- Served as **Services** Coordinator at Mood Indigo, **Operations** Coordinator at E-Cell, and **Infrastructure** Coordinator at Techfest.